T.C.

MARMARA UNIVERSITY

FACULTY OF ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

CSE4078 Introduction to Natural Language Processing Project

Downstream Task Evaluation Report

Project Task

Conversation

Group#7 Members

Mustafa Özgür Hocaoğlu - 150120058
Osman Buğra Göktaş - 150119565
Yiğit Mesut Ak - 150120006
Emir Büçkün - 150119024
Engin Bilici - 150121826

June 1, 2024

Introduction

In this report, we present the evaluation of our base and fine-tuned models using downstream evaluation metrics. We build upon the metrics used in Delivery#3 and explore additional methods to gain a deeper understanding of the model performance.

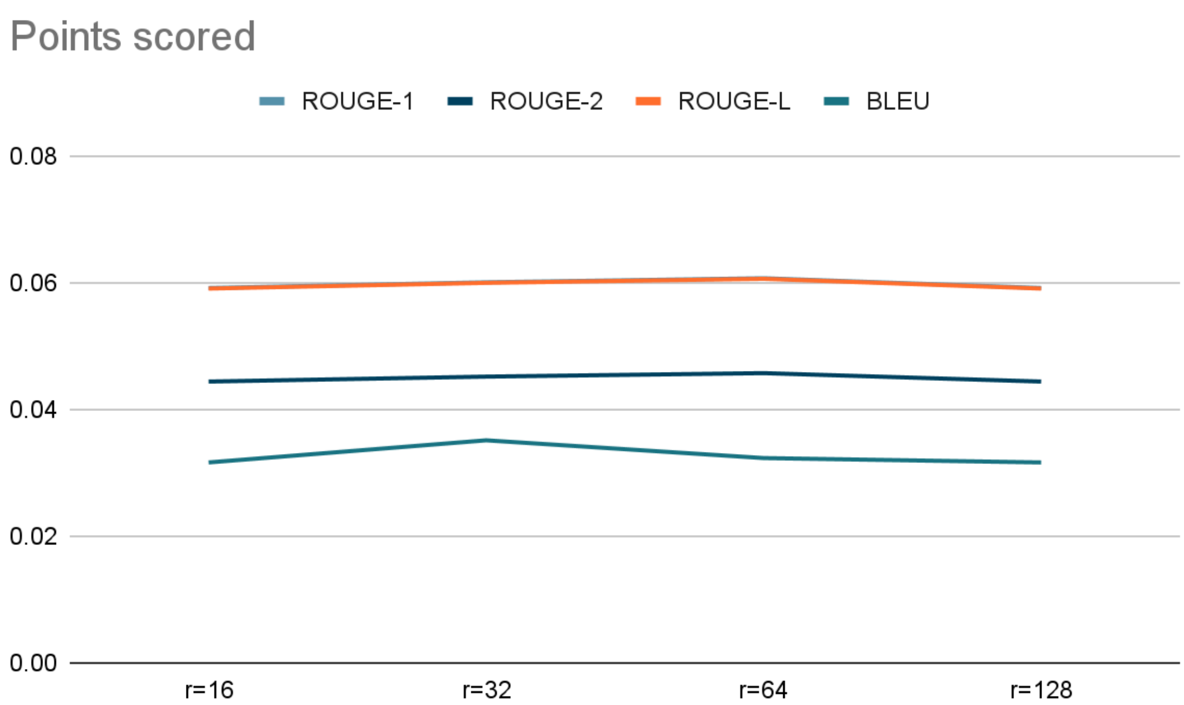
Zero Shot Normal

rank	ROUGE1	ROUGE2	ROUGE-L	BLEU
r = 16	Avg = 0,103313	Avg = 0,091199	Avg = 0,103313	Avg = 0,119126
	Std = 0,09676	Std = 0,095365	Std = 0,09676	Std = 0,202997
r = 32	Avg = 0,10331	Avg =0,09119	Avg = 0,10331	Avg = 0,11912
	Std = 0,0967	Std = 0,09536	Std = 0,09676	Std =0,20299
r = 64	Avg = 0,10331	Avg = 0,09119	Avg = 0,10331	Avg = 0,11912
	Std = 0,09676	Std = 0,09536	Std = 0,09676	Std = 0,20299
r = 128	Avg = 0,10331	Avg = 0,09119	Avg = 0,10331	Avg =0,11912
	Std = 0,09676	Std = 0,09536	Std = 0,09676	Std = 0,20299

Rank is not affecting the average values and standard deviation in ROUGE1, ROUGE2, ROUGE-L, and BLEU evaluation metrics.

One Shot Results

rank	ROUGE1	ROUGE2	ROUGE-L	BLEU
r = 16	Avg = 0,05915	Avg = 0,04439	Avg = 0,05906	Avg = 0,03163
	Std = 0,05781	Std = 0,05464	Std = 0,05787	Std = 0,10793
r = 32	Avg = 0,06005	Avg = 0,04517	Avg = 0,05996	Avg = 0,03512
	Std = 0,06052	Std = 0,05734	Std = 0,06059	Std = 0,11174
r = 64	Avg = 0,06069	Avg = 0,04572	Avg = 0,06060	Avg = 0,03232
	Std = 0,05770	Std = 0,05457	Std = 0,05777	Std = 0,10790
r = 128	Avg = 0,05915	Avg = 0,04439	Avg = 0,05906	Avg = 0,03163
	Std = 0,05781	Std = 0,05464	Std = 0,05787	Std = 0,10793



As the chart indicates, we could not find distinguishing results for the metrics as we change the r value.